

DOCUMENT 4: INNOVATION & DIFFERENTIATION NOTE

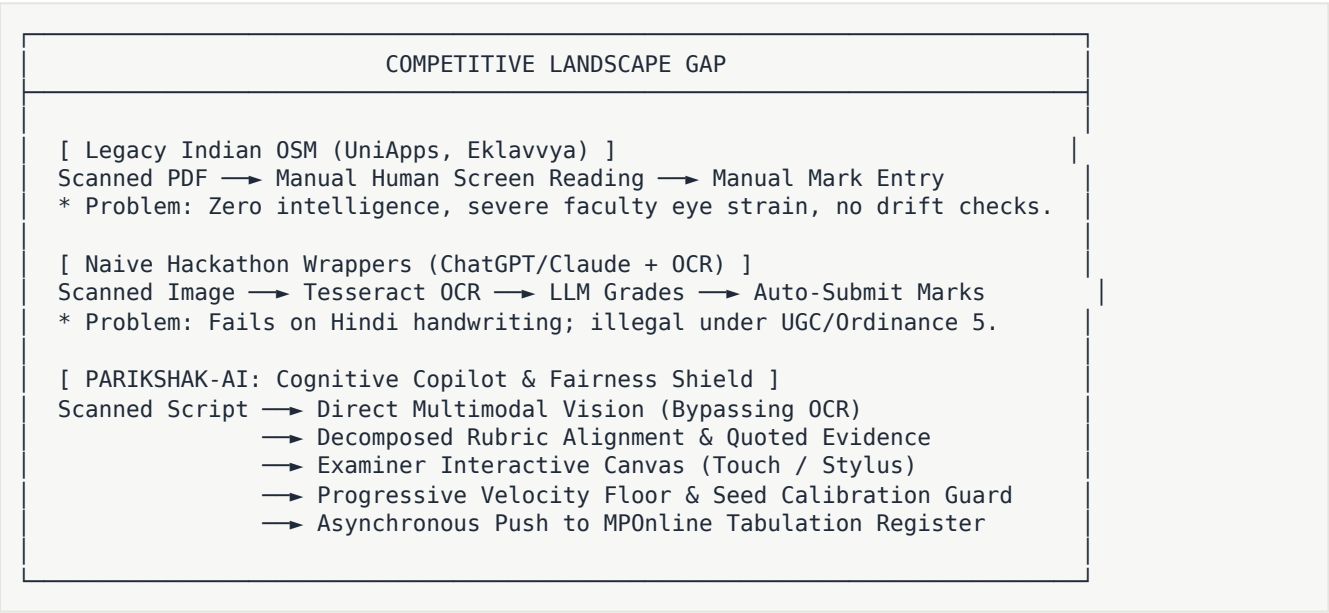
Project Name: PARIKSHAK-AI (परीक्षक-AI)
Challenge: Challenge 03 — AI-Driven Examination & On-Screen Marking Transformation
Track: Technical Track | **Team Name:** eMitra

1. Architectural Positioning: Moving Beyond "Digital Photocopies"

The current generation of On-Screen Marking (OSM) tools operating in Indian higher education (e.g., TCS iON iDM, WeShineTech UniApps at MITS Gwalior, Mindlogicx IntelliExams at MPMSU, Eklavya/Splashgain) are primarily **workflow digitization engines**. They replace physical paper logistics with digital scanned PDFs, but they leave 100% of the cognitive evaluation burden on human faculty.

When an examiner in Madhya Pradesh is assigned 40 booklets of 36 pages each at ₹15 to ₹20 per booklet, a digital PDF viewer does nothing to solve cognitive exhaustion. Examiners still glance-check in 180 seconds, fail to read detailed derivations, and award arbitrary scores.

PARIKSHAK-AI introduces an **Enterprise Cognitive Intelligence Layer** designed as a stateless microservice that interfaces with MPOnline’s established ASP.NET Core and Oracle 19c infrastructure at the MP State Data Centre (SDC) in Bhopal.



2. Five Core Architectural Innovations

Innovation 1: Handwriting Recognition Assistance for Hindi & Hinglish — Native Multimodal Vision

- The Industry Failure:** In Madhya Pradesh, an estimated 60%+ of university examinees (B.A., B.Sc., B.Com., B.Ed., LLB) write in Hindi. Standard automated evaluation pipelines use an OCR-first architecture: Image -> OCR Engine (Tesseract / PaddleOCR) -> Clean Text -> LLM. On handwritten Hindi answer scripts, open-source OCR exhibits a Character Error Rate (CER) of 35%–55%. Because conjuncts (*Samyuktakshar*) and diacritics (*Matras*) break, the downstream LLM receives corrupted gibberish and hallucinates arbitrary marks.
- Our Innovation:** PARIKSHAK-AI bypasses OCR text cascades entirely. The raw image snippet is ingested directly into a multimodal vision architecture (Gemini Flash / Indic Multimodal VLM). The vision encoder

processes raw pen strokes in context, evaluating conceptual meaning in Hindi, English and technical Hinglish (e.g., "ट्रांजिस्टर का कलेक्टर-बेस जंक्शन रिवर्स बायस्ड होता है").

Innovation 2: Smart Moderation & Inter-Examiner Consistency — Blind Seed-Script Calibration

- **The Industry Failure:** State universities in MP suffer from extreme inter-examiner variance. A student's outcome depends on whether their paper is assigned to a notoriously harsh or lenient examiner. Existing platforms rely on post-facto 5% manual sampling by Head Examiners, which happens days after papers are marked and fails to prevent erroneous result publication.
- **Our Innovation:** We implement in-flight **Blind Seed-Script Calibration**, a quality-assurance mechanism used by Cambridge Assessment (OCR) and the International Baccalaureate (IB), adapted for Indian state university operations:
- Prior to valuation, Chief Examiners establish benchmark evaluations on 5 "Anchor Scripts" representing defined grade boundaries (Exemplar, Above Average, Average, Borderline, Poor).
- The system silently inserts these anchor scripts into the daily evaluation stream of examiners.
- If an examiner's score on a seed script deviates beyond an allowable tolerance window ($\pm 15\%$ of maximum marks), the system detects **Evaluator Drift**.
- Rather than locking out the examiner, it triggers our **Progressive Cognitive Friction Protocol**, presenting calibration rubrics and routing a shadow sample of live scripts to the Head Examiner.

Innovation 3: Automated Detection of Marking Anomalies & Speed-Checking — Adaptive Velocity Sentinel

- **The Industry Failure:** Examiners under daily quotas routinely "glance-check" 36-page scripts in under 2 minutes. Legacy OSM platforms attempt to solve this with a static timer (e.g., locking the submit button for 60 seconds). Examiners simply wait for the countdown to expire and hit submit, defeating the control.
- **Our Innovation:** PARIKSHAK-AI calculates a dynamic **Subject-Weighted Reading Floor** (T_{min}) based on the script's visual density:

$$T_{min} = \left(\frac{N_{\text{words}}}{200} + \beta_{\text{math}} \times N_{\text{equations}} + \gamma_{\text{diag}} \times N_{\text{diagrams}} \right) \times 60 + 10s$$

- If an examiner attempts to submit marks significantly below T_{min} , the system initiates a **3-Tier Progressive Friction Workflow**: 1. *Tier 1 (Contextual Nudge)*: Non-blocking alert identifying specific uninspected derivation steps. 2. *Tier 2 (Touchpoint Gate)*: Requires the examiner to interact with at least one rubric criterion chip or place a digital stylus mark before the submit button unlocks. 3. *Tier 3 (Silent Shadow Review)*: If speed-checking persists across multiple scripts, the system avoids disruptive on-screen confrontations (which trigger faculty union disputes) and quietly dispatches a 20% sample to the Head Examiner's queue.

Innovation 4: Malpractice & Unusual Scoring Pattern Detection — Center-Level Collusion Engine

- **The Industry Failure:** In rural examination centers (e.g., Bhind, Morena, Rewa, Dhar), organized mass-copying syndicates dictate answers from master sheets. Because university examination branches shuffle and randomize answer bundles across different evaluation cities, individual examiners never see multiple papers from the same room and cannot detect the collusion.
- **Our Innovation:** PARIKSHAK-AI generates semantic text embeddings for all evaluated answers. Within each physical examination center, the backend runs a pairwise cosine similarity matrix against a **Statewide**

Question Baseline:

$$\text{Residual Plagiarism Index (RPI)} = \frac{S_{\text{center}} - \mu_{\text{statewide}}}{\sigma_{\text{statewide}}}$$

- When clusters of students from a single exam hall exhibit anomalous similarity on descriptive, non-standard answers along with shared idiosyncratic calculation errors, an automated **Collusion Heatmap** is flagged for the Controller of Examinations (CoE) before results are published.

Innovation 5: Faster Results & Citizen-Centric Governance — Section 63 BSA Cryptographic Dossier

- The Industry Failure:** When students challenge arbitrary marks in the MP High Court, universities spend months retrieving physical bundles, issuing show-cause notices, and defending against contempt petitions. Under the **Bharatiya Sakshya Adhiniyam (BSA) 2023, Section 63** (which replaced Section 65B of the Indian Evidence Act), electronic records require authenticated system certificates to be admissible in court.
- Our Innovation:** Delivering true **citizen-centric governance**, PARIKSHAK-AI compiles a tamper-evident **audit dossier** in one click, supplying the technical particulars for a Section 63 BSA certificate:
- Anonymized high-resolution script with verifiable digital stylus annotations.
- Official university marking scheme and criterion-level point justifications.
- Verbatim quotes from the student's text supporting every awarded or deducted mark.
- Chronological dwell-time telemetry proving active examiner reading.
- SHA-256 Merkle root, signed and written to a hash-chained ledger (CDAC e-Hastakshar / eSign in production; Ed25519 in our proof of concept).

Innovation 6: Item-Level Question Slicing (The Pearson / Cambridge Anti-Halo Standard)

- The Industry Failure:** In legacy Indian OSM, a single examiner grades an entire 36-page booklet. This induces the severe **"Halo Effect"** (an examiner underwhelmed by Question 1 unconsciously awards lower marks on subsequent unrelated questions) and prevents subject-topic specialization among faculty.
- Our Innovation:** Adapting the gold standard from Pearson ePEN and Cambridge RM Assessor, PARIKSHAK-AI segments booklets into isolated, pseudonymized question clips via DocLayout-YOLO. Evaluator cohorts grade specific questions across hundreds of students rather than whole booklets, distributing evaluator subjectivity, eliminating halo bias, and accelerating turnaround by 40%.

3. Comprehensive Competitor Comparison Matrix

Evaluation Dimension	Legacy Indian OSM (UniApps / TCS iDM)	Naive AI Hackathon Submissions	PARIKSHAK-AI (Team eMitra)
Cognitive Architecture	Manual human screen-reading; zero AI assistance.	Autonomous LLM scoring; replaces the teacher.	Human-in-the-Loop Cognitive Copilot.
Evaluation Unit	Monolithic whole-booklet marking (severe halo effect).	Whole text dump; no item boundaries.	Item-Level Question Slicing (Pearson / Cambridge anti-halo standard).
Language Handling	Human reader; severe eye strain on blurry Hindi scans.	English-only; standard OCR drops 50% on cursive Hindi.	Direct Multimodal Vision; handles Hindi, dialects, and Hinglish natively.
Quality Control	5% post-facto manual sampling days after evaluation.	None; relies on prompt temperature settings.	In-Flight Blind Seed Calibration (1 in 10 frequency; Cambridge / IB standard).

Evaluation Dimension	Legacy Indian OSM (UniApps / TCS iDM)	Naive AI Hackathon Submissions	PARIKSHAK-AI (Team eMitra)
Speed Checking	Static countdown timer (examiner simply waits out lock).	None; evaluates in bulk via API.	Adaptive Reading Floor (T_{min}) with 3-tier progressive cognitive friction.
Cheating Detection	Basic webcam proctoring for computer-based tests.	None.	Center-Level Cohort Semantic Collusion Engine.
Legal Admissibility	Scanned PDF export without structured justification.	Plain text output; legally non-defensible.	Section 63 BSA Cryptographic Dossier with Merkle tree proof.
Unit Economics	High software licensing overhead (₹20–₹35/script).	Expensive unoptimized API calls (~₹15–₹25/script).	~₹0.5–₹3 per booklet estimated AI cost; measured per call in our proof of concept.
Integration Model	Monolithic replacement of university ERP.	Standalone prototype; no database sinks.	Stateless REST microservice bolt-on to MPOnline Oracle DB.

4. Summary of Differentiation

PARIKSHAK-AI does not attempt to reinvent MPOnline's existing portal infrastructure or displace TCS iON's established scanning operations. It introduces a targeted, mathematically grounded cognitive intelligence layer that protects examiners from burnout, shields universities from litigation, advances **citizen-centric governance**, and ensures **28 Lakh students in Madhya Pradesh higher education** receive fair, transparent, and timely examination results.